Aida Khaleghi

aiida.khaleghi@gmail.com — Tehran, Iran
GitHub — LinkedIn — ORCID — Personal Website

Education

M.Sc. in Artificial Intelligence and Robotics Sep 2024 – Apr 2027
Iran University of Science and Technology, Tehran, Iran
- Thesis (in progress): "Deep Unfolding-Based Anomaly Detection in Medical Imaging"
- Supervisors: Prof. Behrouz Minaei-Bidgoli and Dr. Maryam Amoozegar
- GPA: 19.05/20 (4.0/4.0)

B.Sc. in Computer Engineering Sep 2020 – Jul 2024
Shahid Bahonar University of Kerman, Kerman, Iran
- Project: "Heterogeneous Graph Neural Network for Persian Sentiment Analysis" (grade: 20/20)
- Supervisor: Prof. Mahdi Eftekhari
- GPA: 19.04/20 (3.98/4.0)

Research Interests

- **Machine Learning for Computational Biology**
- **Perturbation Response Modeling**
- **Single-Cell and Spatial Omics**
- **Explainable Models of Biological Systems**

Research Experience

Research Assistant Oct 2025 – Present
Kerman Graduate University of Technology, Kerman, Iran
Under the supervision of Dr. Maryam Amoozegar
- Extended IFISTA-ROTAB, an online tensor-based moving object segmentation method, into a hybrid network
- Replaced temporal and foreground updates with ConvLSTM and CNN modules, retaining recursive least squares (RLS) updates of the background factors
- Implemented and trained a fully online, unsupervised model with a redesigned, more informative loss function
- Improved F-measure by at least 0.1 over IFISTA-ROTAB on LASIESTA and CDNet2014 datasets

Graduate Researcher Jan 2025 – Present
Data Mining Lab, Iran University of Science and Technology, Tehran, Iran
Under the supervision of Prof. Behrouz Minaei-Bidgoli and Dr. Maryam Amoozegar
- Developing an interpretable RPCA-based deep unfolding network for medical image anomaly detection
- Designing an architecture combining convolutional and attention modules to unfold the RPCA optimization
- Implementing the method in PyTorch and evaluating on medical imaging datasets like BraTS (ongoing)

Undergraduate Researcher Feb 2024 – Jul 2024
Shahid Bahonar University of Kerman, Kerman, Iran
Under the supervision of Prof. Mahdi Eftekhari
- Adapted a heterogeneous graph neural network method for sentiment analysis from English to Persian
- Rebuilt the Persian text-to-graph pipeline, modeling reviews as graphs of words, tags, and dependencies
- Trained and evaluated the heterogeneous graph attention network on the Digikala and SnappFood datasets

Publications

- **A. Khaleghi** and M. Amoozegar, "Online Unsupervised Hybrid Network for Moving Object Segmentation in Videos," *manuscript in preparation*, 2026.
- A. Ghasemi, B. Minaei-Bidgoli, and **A. Khaleghi**, "Universal Noise Augmentation (UNA): A Language-Agnostic Framework for Robust Multilingual Text Classification," *under review* at *Machine Learning with Applications*, 2026. Preprint available on SSRN.

Selected Research Projects

- **Double Gene Perturbation Prediction Benchmark:** Benchmarked a CPA-style conditional VAE against additive and mean baselines on Perturb-seq data, using 5 seeds and bootstrap confidence intervals. The additive baseline performed best (MSE 0.045 vs. 0.057). Showed that the CPA variant fails to generalize to perturbations unlike those seen in training.

- **Drug–Drug Interaction Prediction with GNNs:** Built a GraphSAGE link-prediction model with an MLP pair scorer and compared BCE/AUC objectives. Achieved test Hits@20 of 0.896 on 500 drugs and 0.845 on 2,134 drugs without retuning.
- **Multi-Task Breast Ultrasound Analysis:** Extended a MONAI U-Net for joint tumor segmentation and benign/malignant classification on BUS-BRA (1,875 images). Achieved 0.82 Dice and 0.84 ROC-AUC on a held-out test set.
- **Instance Segmentation under Adverse Weather:** Built a 3,704-frame CARLA dataset of clear, rain, fog, and night scenes to test generalization across weather and lighting. Compared YOLOv11n-seg and YOLOv11m-seg, achieving a 4.1× inference speedup with FP16 TensorRT (11.9 to 2.9 ms).

Work Experience

AI Engineer Aug 2025 – Dec 2025
Vida Medicals, Canada (remote)
- Improved mammography asymmetry classification from 60% to 82% using transfer and multi-task learning
- Developed nnU-Net and YOLO models for breast tumor segmentation, achieving 0.83 Dice
- Built a Streamlit annotation tool enabling radiologists to label mammograms using a shared protocol

Teaching Experience

Graduate Teaching Assistant Sep 2025 – Present
School of Computer Engineering, Iran University of Science and Technology, Tehran, Iran
- **Courses (25+ students per course):**
  Fall 2025: Natural Language Processing, Machine Learning, Neural Networks, Deep Learning
  Spring 2026: Advanced Topics in Computer Vision
  Fall 2026: Deep Learning, Applied Artificial Intelligence (for students from other departments)
- Designed programming assignments and problem sets, and graded them with written feedback
- Mentored six graduate students on research projects, from paper selection to evaluation
- Guided an ML project resulting in a co-authored paper (UNA), currently under review
- Delivered a hands-on PEFT workshop and designed, proctored, and graded the NLP practical exam

Undergraduate Teaching Assistant Jan 2023 – Jul 2024
Department of Computer Engineering, Shahid Bahonar University of Kerman, Kerman, Iran
- **Courses (50+ students per course):**
  Spring 2023: Operating Systems (sole TA), Design of Algorithms
  Spring 2024: Basics of Speech and Language Processing (sole TA)
- Designed and graded assignments for all three courses
- Delivered a Transformer architecture workshop for the speech and language processing course
- Led midterm review and problem-solving sessions for algorithms and operating systems

Awards and Honors

- **Exceptional Talent Award, IUST** 2026
  Awarded to the top 5% of students in each faculty per year, based on GPA, teaching, and research. Included a cash prize, an educational grant, and research support.
- **Competitive M.Sc. Admission via Exceptional Talent Track** 2024
  Admitted to IUST without an entrance exam based on outstanding academic performance.
- **Ranked 1st in Computer Engineering Class** 2024
  Ranked 1st out of 90+ students. Shahid Bahonar University of Kerman.
- **Top 1% in the National University Entrance Exam** 2020
  Ranked in the top 1% of approximately 135,000 candidates in the Mathematics and Physics track.

Technical Skills

- **Machine Learning:** deep learning, generative models, graph neural networks, computer vision, natural language processing, multi-modal representation learning
- **Programming:** Python, C++, SQL
- **Frameworks:** PyTorch, PyTorch Geometric, Scanpy, AnnData, scikit-learn, TensorFlow/Keras, Hugging Face Transformers, MONAI, OpenCV, Ultralytics YOLO
- **Software & Infrastructure:** Linux, Git/GitHub, Docker, LaTeX
- **Languages:** English (C1), German (currently learning), Persian (native)